Indian bird audio denoising

(B. Tech. Project-2)

A REPORT

Submitted by

Madhur S. Mundra

23ITUOZ066

Taksh N. Lade

23ITUOS063

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Prof. Nikita P. Desai

Dharmsinh Desai University, Nadiad.



Department of Information Technology

Faculty of Technology,

DHARMSINH DESAI UNIVERSITY

NADIAD 387001

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In the process of writing this report, we used the following AI tools and technologies:

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1. Grammarly (Premium version) was used to generate an outline for this paper and to correct errors in spelling, grammar, and mechanics.
2. Zotero (free) was used as a reference management tool to organize sources and generate citations.
3. Power BI (free) was used to analyze raw data, create initial findings, and generate charts for the conclusions section of this report.
4. Canva (paid version) was used to generate slides for the oral presentation from the text of this paper.

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We declare that the dissertation (for B.Tech in Information Technology) titled " title of your project" is our own work being conducted under the guidance and supervision of Prof. X. Y. Surname.

We further declare that to the best of our knowledge, this dissertation does not contain any part of work which has been submitted for the award of any degree either in this University or in any other University without proper citation.

Signature
Madhur S. Mundra

Signature
Taksh N. Lade

CERTIFICATE

This is to certify that this Report of B. Tech. Project2 submitted for partial fulfillment of B. Tech Semester- VII is a record of the work carried out by

1) Madhur S. Mundra

ID No. 23ITUOZ066, B. Tech. Sem – VII (Information Technology):2026-27

2) Taksh N. Lade

ID No. 23ITUOS063, B. Tech. Sem – VII (Information Technology):2026-27

Guide
Prof. Nikita P. Desai
Associate/Assistant Professor,
Department of Information Technology
Dharmsinh Desai University,
Nadiad–387001, INDIA

HoD
Prof. Dr. V. K. .Dabhi
Head, Dept. of Information Technology
Faculty of Technology
Dharmsinh Desai University
Nadiad–387001, INDIA



Department of Information Technology
Faculty of Technology
Dharmsinh Desai University
College Road, Nadiad-387001, INDIA

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Madhur S. Mundra
Dharmsinh Desai University, Nadiad
October, 2026
madhur.mu5@gmail.com

Taksh N. Lade
Dharmsinh Desai University, Nadiad
October, 2026
Takshlade982@gmail.com

Abstract

Indian Bird Audio Denoising: Cleaning Crowdsourced Labels with an Unsupervised BirdNET Audit
Project-2 by Madhur S. Mundra (23ITUOZ066) and Taksh N. Lade (23ITUOS063)
at
Dharmsinh Desai University, October 2026

Automatic identification of bird species from their calls is important for biodiversity monitoring and conservation, particularly in a megadiverse country such as India. Modern systems learn this task from large collections of annotated recordings. iBC53, hosted on Kaggle, contains recordings of 53 Indian bird species arranged in folders named after the species. It is a crowdsourced dataset: each recording is obtained from Xeno-canto and placed in a folder according to its catalogue label. The difficulty is that the label is attached to the whole recording and not to every moment inside it. A recording of a target bird usually also contains long stretches of silence and often the calls of other species or human and mechanical noise. When such recordings are cut into short training segments, many segments inherit a label that they do not deserve; this phenomenon is called label noise. Listening to and correcting thousands of segments by hand requires ornithological expertise and many hours of work, and is therefore not practical. This project follows a data-centric approach: instead of changing the model architecture, we automatically audit and clean the labels. We segment every recording into non-overlapping 5.0 s chunks and use the pre-trained BirdNET-Analyzer V2.4 as an unsupervised auditor to estimate whether the labelled species is present in each chunk. Using fixed thresholds we tag chunks as empty, intruder, ambiguous or concordant, and keep only concordant chunks for training. On the 53-species iBC53 dataset (1,368 recordings, 9,289 chunks) the audit flagged about 17% of chunks as empty and 0.2% as containing a different species. We then compared an acoustic classifier trained on the raw labels (Model A) with the identical classifier trained on the cleaned labels (Model C), evaluating both on the same held-out raw recordings. Model C improved Macro-F1 from 0.3872 to 0.3968 (+0.0096) while training on about 28% fewer chunks, showing that removing label noise helps even when it reduces the data volume. The cleaned metadata and a reproducible loader are released with this report.

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Abbreviations

1. Introduction

1.1 Introduction to the Research Problem

Automatic identification of bird species from their calls is important for biodiversity monitoring and conservation, particularly in a megadiverse country such as India. Modern systems learn this task from large collections of annotated recordings. iBC53, hosted on Kaggle, contains recordings of 53 Indian bird species arranged in folders named after the species. It is a crowdsourced dataset: each recording is obtained from Xeno-canto and placed in a folder according to its catalogue label. The difficulty is that the label is attached to the whole recording and not to every moment inside it. A recording of a target bird usually also contains long stretches of silence and often the calls of other species or human and mechanical noise. When such recordings are cut into short training segments, many segments inherit a label that they do not deserve. This phenomenon is called label noise. A supervised model trained on noisy labels learns incorrect associations and its accuracy drops. Listening to and correcting thousands of segments by hand requires ornithological expertise and many hours of work, and is therefore not practical.

1.2 Motivation for the Research Work

Recent progress in machine learning has shown that improving the data is often more effective than changing the model; this is the data-centric view. At the same time, strong pre-trained bioacoustic models such as BirdNET are freely available and can identify thousands of species. Although BirdNET is normally used to detect birds, it can also be used in reverse as an automatic auditor: if BirdNET does not hear the labelled species in a segment, that label is suspicious. The motivation of this project is to combine these two ideas: use an existing free model as an unsupervised quality-control step to clean the training labels of an Indian bird dataset, and then measure whether this cleaning actually improves a downstream classifier. A second motivation is honesty and reproducibility: instead of claiming an artificial improvement, we report exactly how much cleaning helps on a fixed, leakage-free evaluation.

1.3 Objectives and Scope of the Research Work

The objectives of this work are as follows.

1) Build an automatic pipeline that converts the iBC53 recordings into fixed 5.0 s chunks at two sampling rates (48 kHz and 16 kHz). 2) Use BirdNET-Analyzer V2.4 with fixed thresholds as an unsupervised auditor that assigns every chunk a triage label (empty, intruder, ambiguous, concordant, or unlabeled). 3) Produce a cleaned dataset containing only chunks in which the labelled bird is confirmed, and release it together with a reproducible data loader. 4) Compare a downstream acoustic classifier trained on the raw labels (Model A) against the same classifier trained on the cleaned labels (Model C) on an identical held-out test set, using Macro-F1. 5) Document the limitations of the approach honestly.

The scope of the comparison is limited to the 11 focus species that are both frequent and auditable by BirdNET. This is a data-cleaning (label-denoising) study; it does not perform waveform or audio enhancement, and the audio itself is never modified. Evaluation uses a recording-level held-out test set of raw chunks so that both models face identical conditions, supported by a small 30-clip manual spot-check.

2. Background Theory

2.1 Bird Acoustic Features and Spectrograms

When a bird call is recorded we obtain a waveform that shows air pressure over time, but the waveform alone is a poor input for classification. A spectrogram is obtained by computing the Short-Time Fourier Transform: the signal is split into short overlapping frames, a Fourier transform is computed for each frame, and the magnitude is displayed with time on the horizontal axis, frequency on the vertical axis and intensity as colour. Bird vocalisations appear as clear, structured traces in a spectrogram and are easier to separate than in raw audio. Because human perception of pitch is logarithmic, the frequency axis is

usually mapped to the mel scale, and the logarithm of the mel-scaled energy gives the log-mel spectrogram, the standard image-like input for convolutional neural networks. Mel-Frequency Cepstral Coefficients (MFCCs) are a compact representation derived from the log-mel spectrum using a Discrete Cosine Transform. A set of MFCC coefficients per frame (39 in this work, including their deltas) forms a small feature vector that captures the spectral shape of the sound and is widely used in speech and animal-call classification, including the acoustic branch of the Gavali et al. model that we compare against.

2.2 BirdNET as an Unsupervised Auditor

BirdNET-Analyzer is a pre-trained deep neural network that identifies bird species from sound. It analyses audio in short sliding windows and returns, for each window, a probability for every species in its label set. In this project BirdNET is used in an unusual way, as an automatic quality-control tool rather than as a detector. For each 5.0 s chunk we take sliding 3.0 s windows, obtain the per-species probability vector, and max-pool the vectors so that the chunk receives a single confidence value for its labelled species and for every other species. Three fixed thresholds then define the triage. If the highest probability is below 0.20 the chunk is treated as empty. If a different species dominates with probability at least 0.65 the chunk is an intruder. If the labelled species dominates with probability at least 0.50 the chunk is concordant. Everything else is ambiguous. Taxa for which BirdNET has no label are marked unlabeled. Using a frozen external model with fixed thresholds keeps the audit reproducible and avoids tuning the cleaning step to the evaluation.

3. Review of Literature

3.1 Bird Species Classification with Deep Learning

Early bird-species recognition systems used hand-crafted features such as MFCCs with classical classifiers. Gavali and Banu (2020) applied convolutional neural networks on a GPU platform for bird species recognition. In their 2025 work, Gavali and Banu propose a visual-acoustic fusion framework that combines a Deep Convolutional Neural Network operating on bird images with a Long Short-Term Memory network operating on MFCCs from bird calls, and report around 94% accuracy on the iBC53 dataset. Their contribution is model-centric: the emphasis is on fusing two modalities to raise accuracy, and the dataset is used as provided. Large-scale benchmarks such as BirdSet (2025) have since standardised audio classification for avian bioacoustics and have shown that generalisation is limited by domain shift and by the quality of the annotations. These works motivate treating data quality itself as the object of study.

3.2 Noisy Labels and Data-Centric Machine Learning

Crowdsourced and web-scraped datasets frequently contain label noise because labels are assigned at a coarse granularity, such as a recording or a track. Research in the data-centric paradigm shows that detecting and removing mislabelled examples can improve downstream performance more than architectural changes, especially when the noise rate is high and the classes are imbalanced. Automatic auditing with a strong pre-trained model is a practical way to detect mislabelled audio without expert review. This project applies that principle to Indian bird calls and measures the effect on a downstream classifier, using BirdNET as the auditor and a compact recurrent classifier as the downstream model.

4. Analysis and Findings

Table 1: Summary of the iBC53 dataset, chunking and BirdNET audit.

Item	Value
iBC53 dataset	53 species, 1,368 recordings, 12.68 h
Chunking	5.0 s non-overlapping; 48 kHz and 16 kHz
Total chunks	9,289 (849 zero-padded, 9.1%)
Concordant (kept)	3,553 (38.2%)
Empty	1,575 (17.0%)
Ambiguous	501 (5.4%)
Intruder	22 (0.2%)
Unlabeled (no BirdNET label)	3,638 (39.2%)



Figure 1: (a) Before cleaning: per-species stacked bars showing mixed categories. (b) After cleaning: concordant-only subset, fully green by construction. (Insert figures/noise_summary.png and figures/noise_summary_cleaned.png.)

The first analysis step was to describe the raw dataset. iBC53 contains 1,368 WAV recordings across 53 species with a total duration of about 12.68 hours (mean 33.4 s per file, range 1.0 to 362.8 s). The largest folder, Mystery mystery, contains 443 recordings that carry no species identity.

Chunking produced 9,289 non-overlapping 5.0 s chunks in both the 48 kHz and the 16 kHz streams. 849 chunks (9.1%) came from the tail of a recording and were zero-padded to reach exactly 5.0 s, and the amount of real audio in each chunk was recorded.

The BirdNET audit classified the chunks as follows: unlabeled 3,638 (39.2%), concordant 3,553 (38.2%), empty 1,575 (17.0%), ambiguous 501 (5.4%) and intruder 22 (0.2%). The

large unlabeled fraction is dominated by the Mystery mystery folder (3,033 chunks) and four taxa for which BirdNET V2.4 has no label (Sphenocichla humei, Chloropsis, Argya longirostris and Cyornis magnirostris). When these no-label taxa are excluded, the label-usable data is 62.9% concordant, 27.9% empty, 8.9% ambiguous and 0.4% intruder. This is the honest measure of how noisy the labelled data actually is.

Cleaning, that is keeping only concordant chunks, retained 3,553 of 9,289 chunks (38.2%). The retained set is therefore green by construction: every bar becomes fully concordant because everything else was removed. This is a standard before/after presentation and does not modify the audio in any way.

Per-species rates vary widely and explain where cleaning matters. Pnoepyga pusilla and Cuculus micropterus are more than 92% concordant, whereas Psittacula eupatria and Glaucidium cuculoides fall below 42% concordant, meaning that more than half of their chunks are essentially empty. Nine taxa disappear completely after cleaning, either because they have no BirdNET label or because all their chunks were empty; these are reported as a limitation.

5. Proposed Work

The proposed solution is a five-stage pipeline. (1) Ingestion: scan the folder structure and build a manifest containing file path, species, recording id and duration. (2) Chunking: load each recording as mono, resample to 48 kHz and 16 kHz, and cut non-overlapping 5.0 s windows, discarding residual tails shorter than 2.0 s and zero-padding the rest; chunk identifiers are deterministic. (3) Auditing: run BirdNET-Analyzer V2.4 on the 48 kHz chunks with 3.0 s sliding windows, max-pool the per-species probabilities, and apply the frozen thresholds to assign a triage label. (4) Cleaning: keep only concordant chunks and write the cleaned metadata file iBC53-Cleaned-Metadata.csv. (5) Classification: extract MFCC features from the 16 kHz chunks and train a compact classifier.

All stages are implemented as small, reproducible Python scripts in the src package: ingest.py, chunking.py, audit.py, clean.py, gavali_features.py, gavali_splits.py, gavali_compare.py, ibc53_loader.py and spotcheck.py. Audio processing uses librosa and soundfile, tables are handled with pandas, and the audit uses the birdnet_analyzer package (version 2.4, CPU). The downstream classifier is trained with scikit-learn for a quick local baseline and with PyTorch (an LSTM over MFCC frames) on a Google Colab T4 GPU for the final results. Every stochastic step uses seed 42. The auditor uses audio sampled at 48 kHz while the downstream model uses 16 kHz. The evaluation is leakage-free: splits are made at the recording level, the gold set (gold_test_150.csv) is excluded from every split and asserted to have zero overlap, and the test set consists of raw chunks for both models so that the comparison isolates the effect of cleaning the training data.

Two models were compared with everything identical except the training data: Model A was trained on the raw labelled chunks, and Model C on the cleaned (concordant-only) chunks. Both were evaluated on the same 234 raw held-out test chunks taken from recordings that were never seen during training. Model A obtained a Macro-F1 of 0.3872 and an accuracy of 0.4402; Model C obtained a Macro-F1 of 0.3968 and an accuracy of

0.4530. Cleaning improved Macro-F1 by +0.0096, about one percent, even though Model C trained on 1,837 chunks compared with 2,563 for Model A, that is about 28% less data. This supports the central hypothesis: less-but-cleaner data trains a better classifier than more-but-noisy data.

Table 2: Ablation - raw data (A) versus cleaned data (C) on the same 234 raw test chunks.

Model	Training data	Train chunks	Test chunks	Accuracy	Macro-F1
A (raw)	raw labels	2,563	234	0.4402	0.3872
C (cleaned)	concordant only	1,837	234	0.4530	0.3968
Delta (C - A)				+0.0128	+0.0096

Figure 2: Confusion matrices of Model A (left) and Model C (right). (Insert figures/confusion_gavali_A.png and confusion_gavali_C.png.)

Table 3: Per-species concordant rate for the 11 focus species.

Species	Total chunks	Kept (concordant)	Concordant %
Pnoepyga pusilla	211	197	93%
Cuculus micropterus	403	373	93%
Arborophila torqueola	347	284	82%
Pomatorhinus ruficollis	335	263	79%
Rimator malacoptilus	181	135	75%
Psilopogon lineatus	159	111	70%
Pellorneum ruficeps	811	535	66%
Liocichla phoenicea	201	130	65%
Cyornis unicolor	276	175	63%
Psittacula eupatria	147	60	41%
Glaucidium cuculoides	276	103	37%

Per-species analysis showed that the largest gains came from species with many test samples (Pellorneum ruficeps +0.14 with n=70, Pnoepyga pusilla +0.23), while apparent losses occurred on species with very few test samples (Arborophila torqueola -0.46 with n=13, Glaucidium cuculoides -0.22). These swings are consistent with small-sample variance. The confusion matrices in Figure 2 show where errors move between the two models. A 30-clip manual spot-check (results/spotcheck_30.csv) was designed to confirm that cleaned chunks genuinely contain the target; it is not used for training. Limitations: the improvement is modest and is reported as measured; the test set is noisy for both models, which is fair but limits absolute accuracy; BirdNET provides no label for five taxa,

which were excluded; and the absolute accuracy is much lower than the 94% reported by Gavali et al. because our evaluation is audio-only on a noisy held-out test set with a compact model, whereas their result fuses images with audio and uses a different split, so the two numbers are not directly comparable.

6. Conclusions

This project showed that an unsupervised audit using a pre-trained bioacoustic model can detect a substantial amount of label noise in a crowdsourced Indian bird dataset, and that removing this noise improves a downstream classifier even when it reduces the training data by roughly 28%. The improvement in Macro-F1 was +0.0096 on a leakage-free raw held-out test set. The result is modest but honest, and the audit pipeline, the cleaned metadata and a loader are released for reproducibility. Future work includes adding a second independent auditor to label the taxa that BirdNET cannot identify; manually

correcting a sample of the 501 ambiguous chunks; enlarging the held-out test to stabilise per-species results; applying data augmentation to compensate for the reduced data volume after cleaning; and comparing against the full image-and-audio fusion model of Gavali et al. when its code becomes available.

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Research Paper

A condensed IEEE-format paper based on this work is in preparation. Its core is the ablation of Model A versus Model C (Table 2) together with the confusion matrices (Figure 2) and the before/after audit figure (Figure 1).

Curriculum Vitae

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